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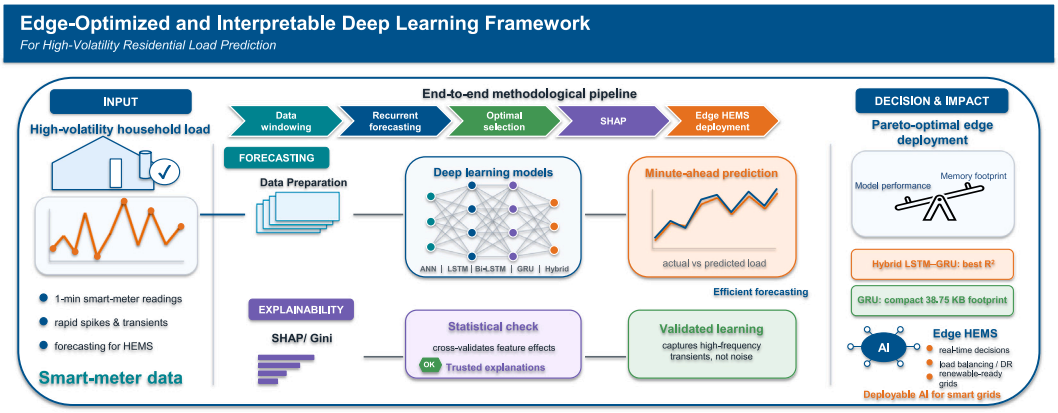
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Edge-optimized and interpretable deep learning framework for high-volatility residential load prediction

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GRAPHICAL ABSTRACT



HIGHLIGHTS

- Dual-validation aligns Kruskal–Wallis and Shapley values for grid transparency.
- Multi-seed tests prove hybrid recurrent networks achieve peak prediction accuracy.
- Zero-lag recurrent models are optimal for edge devices with a 38.75 kilobyte size.
- Explainable methods prove recurrent models exploit minute-level transient signals.
- Edge-optimized frameworks provide a scalable blueprint for volatile global grids.

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ABSTRACT

The transition toward intelligent smart grids requires highly accurate, minute-resolution residential load forecasting. However, deploying deep learning models in real-world home energy management systems is often hindered by strict edge-hardware constraints and a lack of model interpretability. This study proposes a zero-lag, edge-optimized deep learning framework evaluated on high-volatility, minute-resolution smart meter data. A pareto-optimal deployment strategy is established by mathematically quantifying the trade-off between predictive performance, deterministic latency, and static memory footprint. Results from a rigorous 5-run cross-validation protocol show that the proposed hybrid long short-term memory and gated recurrent unit architecture achieves peak accuracy, while the standalone gated recurrent unit model delivers

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statistically equivalent predictive power with a minimal 38.75 kilobyte footprint. To overcome the 'black-box' nature of deep learning, a dual-validation interpretability framework is evaluated. By upgrading standard analysis of variance to the non-parametric Kruskal–Wallis test and cross-examining it against Shapley additive explanations, it is proven that the recurrent architectures achieve superior performance not by memorizing noise, but by dynamically exploiting high-frequency temporal transients. These findings provide a fully reproducible blueprint for deploying interpretable, hardware-efficient artificial intelligence in smart grids.

1. Introduction

The increasing demand for electricity, in conjunction with the need to mitigate carbon emissions, has prompted energy suppliers to enhance the efficiency of their systems. The advancement in technology has resulted in the implementation of smart grids and smart metres, enabling the acquisition of instantaneous data regarding electricity usage patterns [1–3]. Through a comprehensive analysis of the available data, energy suppliers are able to enhance their ability to effectively manage the intricate dynamics of energy supply and demand. This reduces the necessity for costly peak load plants, which are typically deployed during periods of high energy demand [4]. The prediction of residential energy consumption patterns is an essential part of this endeavor and has attracted substantial attention in recent years, as evidenced by the extensive research conducted in this area [5,6].

1.1. Research motivation

The modernization of the electric grid has necessitated a paradigm shift from centralized supply management to decentralized demand-side flexibility. Aggregated load forecasting at the substation level benefits from the smoothing effect of large-scale data. In contrast, single household load prediction presents a complex challenge since individual user behavior introduces volatility and stochastic noise into the data. Despite these challenges, achieving accurate household-level prediction remains the fundamental enabler for effective Demand-side Management (DSM) strategies [7]. Unlike aggregated data, granular household forecasts allow for the precise implementation of Demand Response (DR) programs, empowering consumers to optimize their electricity usage and reduce costs during peak pricing periods [8,9]. With the rising adoption of Electric Vehicles (EVs), accurate demand prediction becomes more critical for optimizing charge scheduling and facilitating peer-to-peer energy trading among prosumers, while increasing profits and improving grid stability [10].

At the residential level, HEMS depend entirely on minute-resolution forecasts to actively manage smart appliances and local generation/storage [11]. This granularity is critical for integrating intermittent renewable energy sources, such as rooftop solar Photovoltaic (PV), by aligning household demand with periods of high generation [12–14].

The modern smart grids are quickly transforming into massive Cyber-Physical Systems (CPS) networks, turning the HEMSs into a highly vulnerable edge node. Deploying robust forecasting models directly at this edge ensures routine energy management while simultaneously detecting sudden load anomalies to build resilience against active cyber–physical threats. As illustrated in Fig. 1, accurate household-level forecasting is the operational engine for future grid capabilities. It directly enables the active management of Distributed Energy Resources (DERs), including residential solar installations and EV charging infrastructure [15].

However, real-world edge deployment enforces strict hardware limitations. Maintaining historical load values (autoregression) or continuously querying external APIs for exogenous data (e.g., weather) introduces severe memory overhead, non-deterministic latency, and privacy vulnerabilities that fundamentally violate the sub-50 KB Static Random-Access Memory (SRAM) constraints of standard microcontrollers. Therefore, this research deliberately imposes a zero-lag constraint, relying exclusively on instantaneous temporal markers to ensure true edge viability. Furthermore, because the objective is real-time HEMS control rather than macro-level seasonal capacity planning, short-term daily and weekly micro-routines are the dominant predictive drivers. For this specific application, minute-level resolution over 73 days yields over 105,000 distinct temporal samples, providing a highly robust statistical foundation for deep learning convergence on immediate household routines.

1.2. Literature review

The need for capturing complex consumption behaviors made the residential energy forecasting progressed from linear statistical models to sophisticated DL architectures. Earlier, Autoregressive Integrated Moving Average (ARIMA) has been widely adopted for aggregated load forecasting but its linear assumptions limit its ability to capture the non-linear volatility in single-household data [16]. To address these stochastic dynamics, a Kalman filter-based bottom-up approach demonstrates that state–space modeling could better track the rapid transitions in household loads compared to static regression methods [17].

To overcome the limitations of linear models, ML algorithms capable of modeling non-linear relationships have been deployed [18]. Further, ANN models gained prominence for its ability to learn complex mappings from historical data [19,20]. Similarly, Support Vector Regression (SVR) models proved effective for datasets with limited samples [21,22]. However, as summarized in Table 1, these models lack inherent memory mechanisms, making them less effective at capturing long-term temporal dependencies in time-series data [23].

The DL models, particularly Recurrent Neural Networks (RNNs), marked a paradigm shift by introducing memory gates to model sequential dependencies [24,25]. The LSTM network became a standard for energy forecasting due to its ability to mitigate the vanishing gradient problem [26,27]. For instance, LSTM variants have been successfully applied to predicting commercial heating systems [28] and convolutional structures for household power [29]. Other studies have emphasized data pre-analysis [30] and theory-guided ensemble frameworks [31] to further improve accuracy. Specifically for residential load forecasting, LSTM models have shown significant improvements over traditional methods [32]. The GRU network was introduced as a more computationally efficient variant, offering comparable performance with fewer parameters [33,34]. Subsequent applications included stacked GRU-RNN models for renewable smart grid operations [35] and optimization frameworks reducing electricity loads [36]. Further advancements included Bidirectional Long Short-Term Memory (Bi-LSTM) models, which process data in both forward and backward directions [37,38], and Convolutional Neural Networks (CNNs), which treat time-series as spatial data to extract local features [39,40]. Recently, hybrid approaches have combined CNNs with RNNs to leverage both feature extraction and sequential modeling capabilities [41,42]. This includes clonal selection algorithm-based models for commercial buildings [43], integration with ML regressors [44], and image-based multi-step forecasting incorporating phase space reconstruction [45]. This domain continues to expand rapidly; recent deep learning algorithms such as BiLSTM-MHSA and CNN-BiGRU-AM have demonstrated exceptional performance in complex domains like fuel cell degradation prediction [46,47]. However, while Multi-Head Self-Attention (MHSA) and CNN architectures yield exceptional accuracy, their elevated parameter counts and computational complexity currently exceed the strict memory constraints mandated by edge-HEMS application.

Single models often struggle to capture the multi-scale nature of residential consumption, which involves both long-term daily trends and

Nomenclature

$[h_{t-1}, x_t]$	Concatenation of previous hidden state h_{t-1} and input x_t
$\bar{y}$	Mean of actual electricity consumption
$\delta(L)$	Error at the output layer
η	Learning rate
$\hat{y}_i$	Predicted electricity consumption
$\mathcal{O}(\cdot)$	Big O notation for computational complexity
∇Loss	Gradient of the loss function
$\nabla a(L)$	Gradient of the activations at the output layer
$\nabla b(l)$	Gradient of the biases at layer l
$\nabla f(z(l))$	Derivative of the activation function for weighted input of layer l
$\nabla W(l)$	Gradient of the weights at layer l
$\odot$	Element-wise multiplication
$\odot$	Element-wise multiplication
$\sigma(\cdot)$	Sigmoid activation function
$\tilde{h}_t$	Candidate hidden state activation at time step t
a_{0j}	Activation of the jth neuron in the input layer
a_{1j}	Activation of the jth neuron in the hidden layer
a_{2k}	Activation of the kth neuron in the output layer
$b_i, b_f, b_g, b_o, b_z, b_r, b_h$	Bias vectors for the gates
b_{1j}	Bias of the jth neuron in the hidden layer
b_{2k}	Bias of the kth neuron in the output layer
c_t	Cell state at t
$f(\cdot)$	Activation function
f_t	Forget gate activation at t
g_t	Candidate cell state activation at t
H	Hidden size (number of units in the recurrent layer)
h_t	Hidden state at t
h_t^b	Backward hidden state at t
h_t^f	Forward hidden state at t
I	Input sequence vector size
i_t	Input gate activation at t
n	Number of samples
o_t	Output gate activation at t
p-value	Probability value representing statistical significance
R^2	Coefficient of Determination
r_t	Reset gate activation at t
$W(l+1)$	Weight matrix between the $(l+1)$-th and lth layers
$W_i, W_f, W_g, W_o, W_z, W_r, W_h$	Weight matrices for the gates
w_{1ji}	Weight between the ith neuron in the input layer and the jth neuron in the hidden layer
w_{2kj}	Weight between the jth neuron in the hidden layer and the kth neuron in the output layer
x_j	Input feature j
y_i	Actual electricity consumption
z_t	Update gate activation at t
z_{1j}	Weighted sum of the jth neuron in the hidden layer
z_{2k}	Weighted sum of the kth neuron in the output layer

short-term minute-wise fluctuations. This has led to the development of complex hybrid architectures, ranging from baseline hybrid deep learning [48,49] to double-layer bidirectional networks [50] and SVR-LSTM variants [51]. Concurrently, clustering-based approaches [52,53] and autoencoder networks [54] have been utilized to decompose the load profile into distinct components. A combined DL model specifically designed for single-household users shows that explicitly addressing multi-time scale behaviors is essential for accurate forecasting [55]. Further, the integration of feature engineering [56–58] and Explainable Artificial Intelligence (XAI) techniques has become critical in ensuring the transparency and trustworthiness of DL models [59]. The "black-box" nature of DL models remains a barrier to deployment, necessitating interpretability frameworks like SHAP to validate model decisions [60,61].

Attention mechanisms, allows the model to focus on the most relevant historical data points, proving effective for long and complex time series data [62,63]. The success of attention led to the development of the Transformer architecture, which relies entirely on self-attention mechanisms to model global dependencies without the need for recurrent processing [64]. Transformer variants, such as the Informer and Autoformer, have set new state-of-the-art performance ceilings in long-sequence cloud forecasting [65,66]. Recent developments, such as the xLSTM architecture, aim to combine the strengths of both RNNs and Transformers [67]. However, we explicitly define the operational scope of this research as strictly bounded by the physical memory and deterministic latency realities of embedded edge hardware, not cloud-based accuracy maximums. The exclusion of Transformers is a deliberate architectural scoping decision dictated by mathematics. Self-attention mechanisms scale quadratically in memory and compute complexity ($O(N^2)$). Storing Attention weights, Key/Query/Value matrices, and positional encodings requires massive parameter counts (typically in the tens of megabytes), which violently violates the strict SRAM limit of standard microcontrollers like the ESP32. Our framework explicitly fills the critical operational gap at the grid's edge, enabling advanced Artificial Intelligence (AI) control in localized environments where state-of-the-art Transformers physically cannot be deployed.

1.3. Research gap

While DL models demonstrate high accuracy in load forecasting, transition from theoretical research to real-world edge deployment is hindered by three critical gaps.

- The 'black-box' nature of recurrent networks limits their adoption by utility operators who require physically meaningful decision-making. Furthermore, when studies attempt statistical validation, they frequently misapply standard Analysis of Variance (ANOVA) to inherently right-skewed, heteroscedastic residential load data, failing to provide mathematically rigorous proof of feature significance. Existing research rarely bridges the gap between robust, non-parametric statistical baselines and modern XAI.
- The literature frequently ignores the hardware constraints of HEMS. A model with superior accuracy is impractical if its memory footprint, parameter count, and computational Floating Point Operations (FLOPs) exceed the strict SRAM capacity of low-cost, edge-deployed microcontrollers.

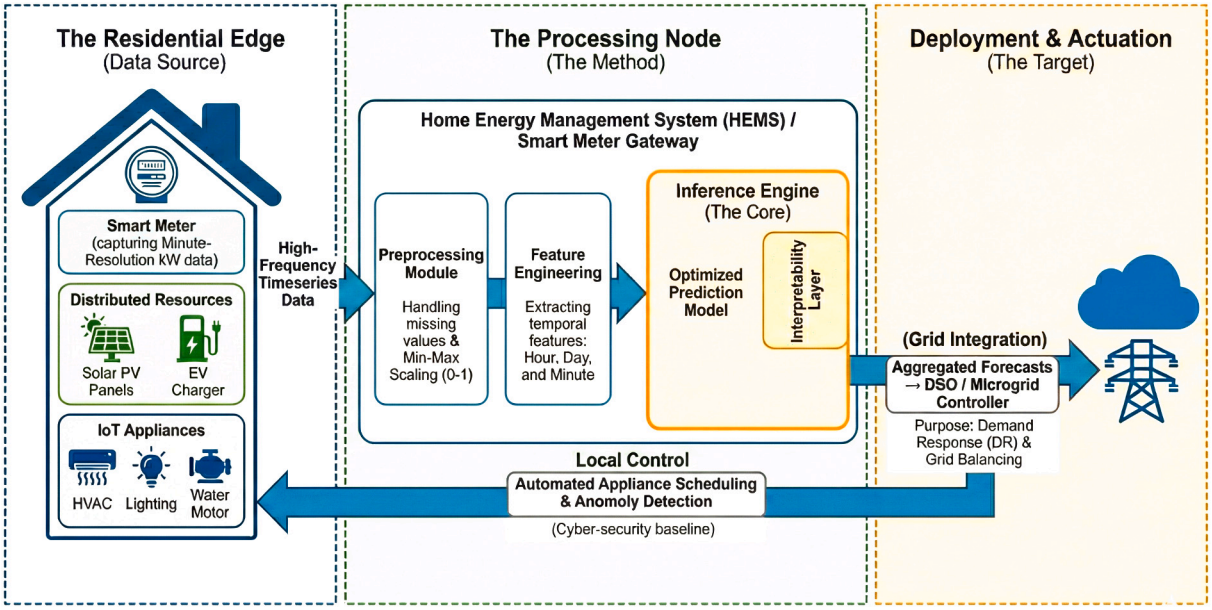


Fig. 1. Edge-centric HEMS deployment architecture.

Table 1
Comparative study of methodologies used for energy consumption prediction models.

Methodology	Approach	Key features	Limitations	Insights
ARIMA [16]	Statistical time series analysis	Captures linear patterns; widely used in energy forecasting	Limited in capturing non-linear patterns; sensitive to outliers	Utilizes autoregressive, integrated, and moving average components to model time series data
ANN [19,20,23]	Machine learning algorithm inspired by the brain	Effective in capturing non-linear patterns	Black-box nature may hinder interpretability of results	Employs feedforward architecture with multiple layers of neurons for learning complex mappings between inputs and outputs
LSTM [24,26–30]	Deep learning model for sequential data analysis	Captures long-term dependencies in time series data	Computational complexity may limit scalability	Utilizes memory cells and gating mechanisms to retain information over long sequences; suitable for time series data
GRU [33–36]	Variant of RNN with reduced parameters	Faster training; efficient in capturing temporal dependencies	Less expressive than LSTM; may not capture long-term dependencies	Employs reset and update gates to control flow of information; balances expressiveness and efficiency
Bi-LSTM [37,38]	Modification of LSTM with bidirectional layers	Incorporates both past and future context for prediction	Increased computational complexity; potential overfitting	Employs forward and backward LSTM layers to capture dependencies in both directions; useful for time series with bidirectional dependencies
CNN [39,40]	Convolutional neural network for spatial analysis	Treats time series data as spatial data	May require substantial tuning; less common in time series analysis	Utilizes convolutional layers with filters to capture spatial–temporal patterns in time series data
Hybrid Models [21,48–54]	Combination of traditional and deep learning	Leverages strengths of both approaches	Complexity in model design and interpretation; data integration	Combines algorithms such as ARIMA, ANN, LSTM, etc., to harness their complementary strengths in forecasting

- The majority of minute-resolution forecasting studies rely on stable, highly predictable datasets. There is a lack of validated models for rapidly developing economies, where grid volatility, frequent power interruptions, and stochastic residential behavior present a challenging forecasting environment.

1.4. Contributions

This study addresses the critical need for standardized evaluation in residential load forecasting by establishing a comprehensive, reproducible benchmark. The primary contributions of this work are:

- A dual-validation interpretability framework is proposed to cross-examine non-parametric statistical significance (Kruskal–Wallis H-Test) against tree-based feature importance (Gini) and deep learning explainability (SHAP), explicitly correcting the distributional flaws of standard ANOVA when applied to right-skewed load data.
- A hybrid recurrent neural network architecture is proposed to effectively capture multi-scale temporal dependencies in high-frequency data, achieving superior predictive performance compared to highly optimized traditional machine learning baselines.
- A rigorous 5-run multi-seed cross-validation is conducted to establish statistical stability bounds, mathematically proving that the

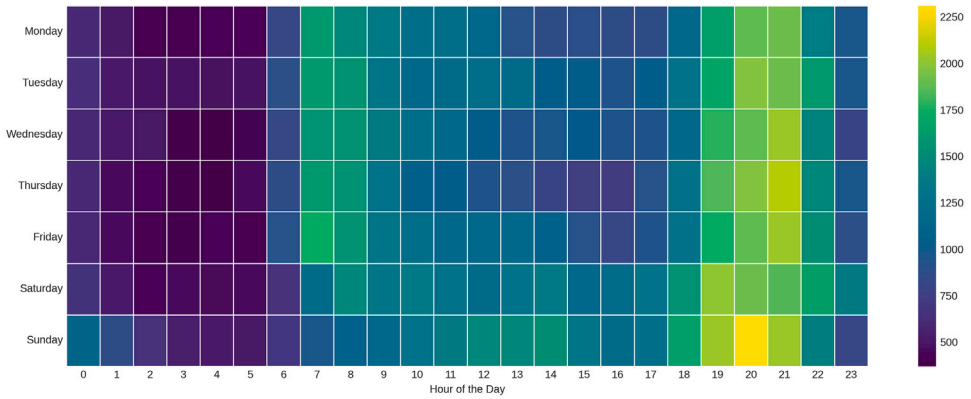


Fig. 2. Weekly consumption pattern.

predictive capabilities of the deep learning models are generalized rather than artifacts of random noise memorization.

- A pareto-optimal deployment strategy is established for edge devices by mathematically quantifying the trade-off between predictive performance, deterministic inference latency, and static SRAM memory footprint, explicitly identifying the zero-lag GRU as the optimal hardware-constrained solution.

2. Data description and handling

The Indian Dataset for Ambient Water and Energy (IAWE) dataset obtained from ground floor and first floor of a residential building has been used in this work [68]. The data on energy consumption was recorded over a span of seventy-three consecutive days, with various household appliances including water motors, air conditioners, laptops and refrigerators. The data involving energy consumption from both floors were integrated.

A multi-faceted view of usage patterns in the dataset is presented in Figs. 2 and 3. The weekly consumption distribution using a heatmap is visualized in Fig. 2 reveals recurring patterns and potential anomalies in energy usage across different days and times. The average hourly consumption trend across the entire dataset presented in Fig. 3 offer insights into consistent daily usage patterns.

The weekdays and weekends of consumption are represented by the independent variable "days". Similarly, the time of consumption is represented by the independent variables "hours" and "minutes". The total power consumption becomes the dependent variable. The proposed prediction model has three independent variables: days, hours, and minutes, and one dependent variable: total power consumption, as shown in Fig. 4.

Preprocessing is done to enhance the usability of the dataset for subsequent analysis by removal of missing data, conversion of the time interval from its original unit of seconds to minutes. The data was scaled to the range of 0 to 1 using min–max scaling. Following the preprocessing stage, the dataset was partitioned using a strictly chronological 90/10 train-test split. The initial 90% of the timeline was used sequentially for training and validation, while the final 10% was strictly held out as an unseen future test set. This deterministic, non-random splitting strategy guarantees zero future-to-past data leakage, which is critically required for valid time-series forecasting. The training process involved constant evaluation of the model for each epoch, with the loss being calculated at each step.

3. Methodology

3.1. Problem formulation and edge-hardware constraints

The forecasting task is formulated as a high-frequency temporal profile learning problem rather than traditional lag-based sequence forecasting. The input sequence is strictly defined by instantaneous temporal markers at time t (zero-lag), and the forecasting horizon is an immediate instantaneous mapping of power consumption.

This deliberate exclusion of lagged historical values and exogenous API data is dictated by strict, quantifiable mathematical constraints on the edge microcontroller's SRAM (typically ~520 KB for an ESP32). Furthermore, incorporating exogenous inputs (such as external API weather queries) forces the microcontroller to maintain an active WiFi/TLS stack, which consumes 80 to 120 KB of runtime SRAM before inference even begins. Therefore, a purely temporal, zero-lag profile is mandatory to ensure the deep learning framework remains operationally viable for this class of embedded edge devices.

A total of four features, including power consumption, were extracted from the preprocessed data as explained in the previous section. Five DL models, including four standalone models and a proposed hybrid LSTM - GRU model, and six traditional ML models were trained on the preprocessed data set.

3.2. DL models

The deep learning architectures were deliberately selected to represent the spectrum of sequential modeling capabilities versus computational cost. Feedforward ANNs serve as the non-sequential baseline. LSTM and GRU architectures represent the state-of-the-art for resource-constrained time-series forecasting; unlike computationally heavy Transformer models which are unsuited for edge HEMS hardware, RNN variants offer a realistic pathway to deployment. The Hybrid LSTM-GRU was specifically hypothesized to test whether combining long-term memory gates with efficient short-term update gates could uniquely capture multi-scale volatility.

3.2.1. ANN

The ANN algorithm represents a foundational feedforward neural network architecture. A three-layer dense architecture was implemented comprising 128, 64, and 32 neurons in successive hidden layers, following a tapering design principle that progressively compresses feature representations toward the final regression output. The model employs Rectified Linear Unit (ReLU) activation functions throughout the hidden layers to introduce non-linearity while mitigating vanishing gradient issues common in deep architectures.

The mathematical formulation of the feedforward ANN with three hidden layers follows standard backpropagation principles:

Forward Propagation:

Input Layer:

$$a_{0j} = x_j \quad \text{for} \quad j = 1, 2, \ldots, n \tag{1}$$

Hidden Layer 1 (128 neurons):

$$z_{1j} = b_{1j} + \sum_{i=1}^{n} w_{1ji} \cdot a_{0i} \tag{2}$$

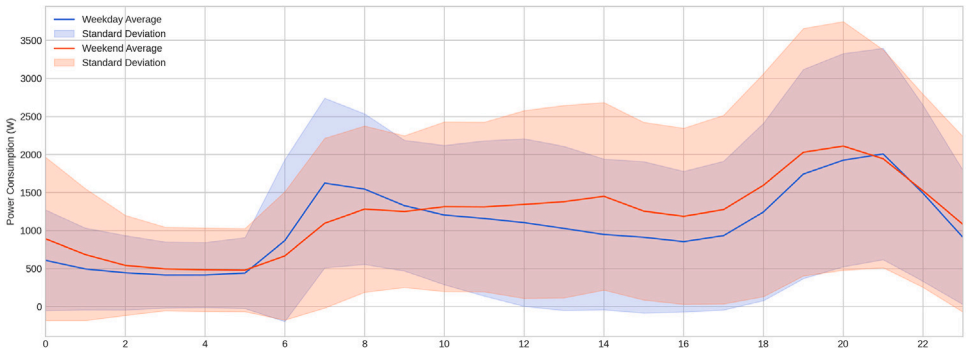


Fig. 3. Average hourly trend.

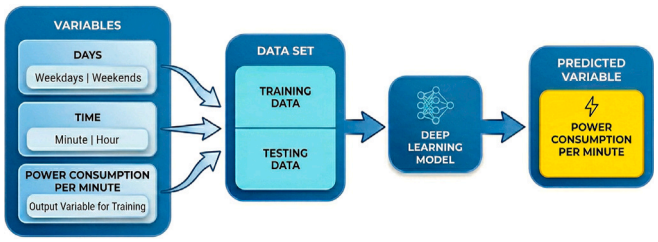


Fig. 4. Architectural flow and feature extraction process of the proposed zero-lag residential load prediction framework.

$$a_{1j} = \text{ReLU}(z_{1j}) = \max(0, z_{1j}) \tag{3}$$

Hidden Layer 2 (64 neurons):

$$z_{2j} = b_{2j} + \sum_{i=1}^{128} w_{2ji} \cdot a_{1i} \tag{4}$$

$$a_{2j} = \text{ReLU}(z_{2j}) \tag{5}$$

Hidden Layer 3 (32 neurons):

$$z_{3j} = b_{3j} + \sum_{i=1}^{64} w_{3ji} \cdot a_{2i} \tag{6}$$

$$a_{3j} = \text{ReLU}(z_{3j}) \tag{7}$$

Output Layer:

$$z_4 = b_4 + \sum_{j=1}^{32} w_{4j} \cdot a_{3j} \tag{8}$$

$$\hat{y} = z_4 \tag{9}$$

Backpropagation:
Error at the output layer:

$$\delta(L) = \frac{\nabla \text{Loss}}{\nabla a(L)} \tag{10}$$

Error propagation through hidden layers:

$$\delta(l) = (W(l+1)^T \cdot \delta(l+1)) \odot \nabla \text{ReLU}(z(l)) \tag{11}$$

where $\nabla \text{ReLU}(z) = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{otherwise} \end{cases}$

Gradient computation for weights and biases:

$$\nabla W(l) = \delta(l) \cdot a(l-1)^T \tag{12}$$

$$\nabla b(l) = \delta(l) \tag{13}$$

Parameter updates via gradient descent:

$$W(l) = W(l) - \eta \cdot \nabla W(l) \tag{14}$$

$$b(l) = b(l) - \eta \cdot \nabla b(l) \tag{15}$$

where η represents the learning rate, controlling the step size during parameter updates.

3.2.2. LSTM network

The LSTM neural network has four main gates, namely, cell, forget, input, and output. The cell gate remembers and forgets details as per the context of the training data. The multiplication operation is performed initially for forgetting and remembering some information, later, the addition operation is performed for some additional information. The forget gate uses a sigmoid activation function, which varies smoothly between 0 to 1. If the information is found similar, it will have more 1s in the vector, and if the information is not similar, it will have more 0s. Additional information can also be added through the input gate, which involves a three-step process. First, adding the values to the cell state is done by the sigmoid function. Then, the tanh function is used to construct a vector using the values added in cell gate. Finally, the sigmoid gate value is multiplied by the tanh function, and the addition operation is used to add these helpful values to the cell state. The equations below outline the calculations involved in each component:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{16}$$

The sigmoid function is used to squash the gate values between 0 and 1.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{17}$$

Similarly, the forget gate calculates the extent to which the LSTM should forget the last cell state.

$$g_t = \tanh(W_g \cdot [h_{t-1}, x_t] + b_g) \tag{18}$$

The cell state computes a new candidate value for the cell state using the hyperbolic tangent (tanh) activation function.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{19}$$

The output gate determines the extent of the current cell state that should be outputted. The current cell state (c_t) is upgraded by:

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t \tag{20}$$

$$h_t = o_t \odot \tanh(c_t) \tag{21}$$

3.2.3. Bi-LSTM network

Bi-LSTM is an expansion of the LSTM that allows information to be passed via the LSTM network in both directions. Bi-LSTM does not do the inverse of LSTM; it only adds an extra layer of LSTM in the opposite direction. The output of both LSTM Layers is combined to create the final result.

Forward LSTM:

$$i_t^f = \sigma(W_i^f \cdot [h_{t-1}^f, x_t] + b_i^f) \quad (22)$$

$$f_t^f = \sigma(W_f^f \cdot [h_{t-1}^f, x_t] + b_f^f) \quad (23)$$

$$g_t^f = \tanh(W_g^f \cdot [h_{t-1}^f, x_t] + b_g^f) \quad (24)$$

$$o_t^f = \sigma(W_o^f \cdot [h_{t-1}^f, x_t] + b_o^f) \quad (25)$$

$$c_t^f = f_t^f \odot c_{t-1}^f + i_t^f \odot g_t^f \quad (26)$$

$$h_t^f = o_t^f \odot \tanh(c_t^f) \quad (27)$$

Backward LSTM:

$$i_t^b = \sigma(W_i^b \cdot [h_{t+1}^b, x_t] + b_i^b) \quad (28)$$

$$f_t^b = \sigma(W_f^b \cdot [h_{t+1}^b, x_t] + b_f^b) \quad (29)$$

$$g_t^b = \tanh(W_g^b \cdot [h_{t+1}^b, x_t] + b_g^b) \quad (30)$$

$$o_t^b = \sigma(W_o^b \cdot [h_{t+1}^b, x_t] + b_o^b) \quad (31)$$

$$c_t^b = f_t^b \odot c_{t+1}^b + i_t^b \odot g_t^b \quad (32)$$

$$h_t^b = o_t^b \odot \tanh(c_t^b) \quad (33)$$

Final Output:

$$h_t = [h_t^f, h_t^b] \quad (34)$$

The h_t is obtained by concatenating the forward and backward hidden states at each time step.

3.2.4. GRU network

The GRU algorithm is a variation of the LSTM algorithm, where the forget and input gates are integrated into one update gate. The update gate decides which information should be kept, forgotten or added to the memory cell. It operates in the same way as the LSTM cell, but with fewer parameters. The reset gate regulates the amount of information saved in the previous state, and the new state is determined by merging the reset and candidate states. The candidate state is computed using the tanh function applied to the weighted sum of the previous state and the input data.

Below equations describe the calculations involved in the GRU algorithm, enabling the update (z_t), reset (r_t), and hidden (h_t) states to be determined and updated throughout the RNN.

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \quad (35)$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r) \quad (36)$$

$$\tilde{h}_t = \tanh(W_h \cdot [r_t \odot h_{t-1}, x_t] + b_h) \quad (37)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (38)$$

3.2.5. LSTM–GRU network

The proposed LSTM - GRU hybrid algorithm combines the strengths of both LSTM and GRU models to improve the performance of energy consumption prediction. It incorporates the long-term memory retention capability of LSTM and the computational efficiency of GRU.

The initial LSTM layer, with its dedicated cell state (c_t) and explicit forget gate (f_t), is powerful at capturing longer-term dependencies and mitigating the vanishing gradient problem over extended sequences [27]. The subsequent GRU layer has a simpler architecture with an integrated update gate (z_t) that combines the forget and input functions. It is computationally efficient and has been shown to be highly effective at capturing more recent, shorter-term temporal patterns [42,49]. By stacking the GRU layer on top of the LSTM layer, the model aims to create a hierarchical representation of temporal features: the LSTM layer learns a robust long-term context, which is then refined by the GRU layer with a focus on more immediate sequential information.

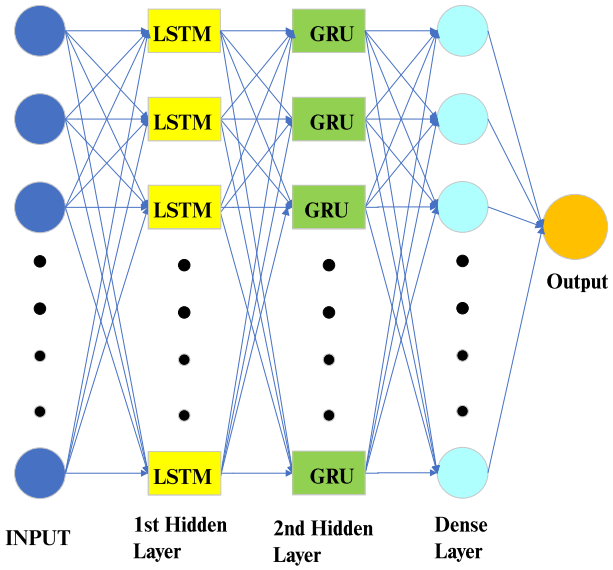


Fig. 5. LSTM + GRU hybrid algorithm.

The LSTM and GRU layers are placed on top of one another to create a multi-layer RNN in this hybrid model. The LSTM layer captures long-term dependencies and retains memory of past information, while the GRU layer focuses on capturing short-term patterns and adapting to the current input. This combination allows the model to effectively capture both long-term and short-term dynamics in the dataset.

The LSTM layer in the hybrid model consists of four gates, similar to the standalone LSTM algorithm. It retains and forgets information based on the context of the training data, ensuring the preservation of relevant long-term patterns. On the other hand, the GRU layer introduces the update gate that determines the information to be kept, forgotten, or added to the memory cell. By combining the strengths of both LSTM and GRU, the hybrid model achieves a balance between memory retention and computational efficiency.

During training, the hybrid model learns to optimize the weights and biases of the LSTM and GRU layers using backpropagation and gradient descent algorithms. The model is trained on the preprocessed data, where the features are scaled down using the MinMaxScaler to ensure proper convergence and learning.

The hybrid model combines both the LSTM and GRU layers. As shown in Fig. 5, a stacked architecture is created when the output of the LSTM layer is used as the input for the GRU layer.

LSTM : $i_t, f_t, g_t, o_t, c_t, h_t$

GRU : $z_t, r_t, \tilde{h}_t, h_t$

These equations represent the calculations performed at each timestep of the LSTM and GRU layers in the hybrid model.

3.2.6. Model configuration and hyperparameters

The hyperparameter configurations for the implemented models are detailed in Table 2. A dual optimization strategy was employed to identify optimal model architectures. For recurrent neural networks

Table 2
Hyperparameter configuration for all evaluated DL models.

Hyperparameter	ANN	LSTM	Bi-LSTM	GRU	LSTM+GRU
Input Shape	(3,)	(1, 3)	(1, 3)	(1, 3)	(1, 3)
Hidden Layers	3	2	2	2	2
Layer 1 Type	Dense	LSTM	Bi-LSTM	GRU	LSTM
Layer 1 Units	128	64	32	32	50
Layer 2 Type	Dense	LSTM	Bi-LSTM	GRU	GRU
Layer 2 Units	64	64	64	32	50
Layer 3 Type	Dense	–	–	–	–
Layer 3 Units	32	–	–	–	–
Dropout	–	0.2	0.2	0.2	–
Activation Function	ReLU	tanh	tanh	tanh	tanh
Recurrent Activation	–	sigmoid	sigmoid	sigmoid	sigmoid
Epochs	140	116	116	116	110
Learning Rate	adam	0.001	0.001	0.001	0.001
Batch Size	256	128	128	128	256
Optimizer	Adam	Adam	Adam	Adam	Adam
Loss Function	MSE	MSE	MSE	MSE	MSE
Total Parameters	10,881	50,497	75,393	9921	26,151
Memory Size (KB)	42.50	197.25	294.50	38.75	102.15
Optimization	Grid Search	Grid Search	Grid Search	Grid Search	Grid Search

(LSTM, GRU, Bi-LSTM), systematic hyperparameter tuning was performed using grid search with 3-fold cross-validation. The search space explored layer units ([32, 64]), dropout rates ([0.1, 0.2]), learning rates ([0.001, 0.01]), and batch sizes ([128, 256]), resulting in comprehensive evaluation across 32 parameter combinations per model. For each configuration, the model was trained and validated across three folds, with the best-performing parameters selected based on cross-validated R^2 score.

In contrast, the ANN and Hybrid LSTM-GRU models were adopted after empirical analysis which revealed that simpler architectures achieved superior performance with manually-tuned configurations in addition to grid search-optimized variants. The ANN model employs a 128-64-32 three-layer architecture with ReLU activation and Adam optimizer (default learning rate). The Hybrid model combines LSTM (50 units) and GRU (50 units) layers with a learning rate of 0.001, designed to leverage LSTM's long-term dependency modeling alongside GRU's computational efficiency.

The MSE was selected as the loss function across all models, as it is a standard metric for regression tasks that penalizes larger errors more significantly, driving the model to make more accurate predictions. EarlyStopping with patience = 15 epochs and minimum delta = 0.0001 (starting from epoch 100) was implemented to prevent overfitting while ensuring sufficient training convergence. The actual epoch counts achieved through this mechanism are presented in Table 2.

3.3. Traditional ML models

To establish a comprehensive benchmark, six traditional ML algorithms were systematically selected to represent diverse learning and computational characteristics. While persistence and Seasonal AutoRegressive Integrated Moving Average (SARIMA) models are standard in macro-grid forecasting, minute-level residential data is highly stochastic and non-stationary. Applying SARIMA at a minute resolution requires massive differencing and computational overhead that is fundamentally incompatible with microcontroller execution. Therefore, highly optimized tree-based ML ensembles and linear methods were selected based on (1) algorithmic diversity; (2) documented success in time-series forecasting applications; (3) computational tractability for real-time deployment; and (4) interpretability through feature importance mechanisms [69–72].

3.3.1. Ensemble gradient boosting methods

XGBoost and LightGBM were selected as state-of-the-art gradient boosting implementations. These algorithms have demonstrated superior performance in numerous energy forecasting studies [69]. The XGBoost employs regularized boosting to prevent overfitting and handles sparse data efficiently. The LightGBM utilizes a leaf-wise tree growth strategy and histogram-based learning for enhanced speed and memory efficiency. These algorithms are proven in capturing complex nonlinear relationships and feature interactions which are critical for energy consumption modeling.

3.3.2. Ensemble tree methods

Random Forest and Extra Trees represent bagging-based ensemble approaches that construct multiple decision trees and aggregate their predictions. The Random Forest uses bootstrap sampling and optimized split selection. The Extra Trees introduces additional randomization by selecting splits at random, often improving generalization. Both methods provide inherent resistance to overfitting and offer interpretable feature importance scores, making them popular choices for energy systems applications where model transparency is valued [71].

3.3.3. Single decision tree

A standalone Decision Tree serves as a baseline to quantify the performance gains achieved through ensemble aggregation. While prone to overfitting when grown deeply, decision trees offer unparalleled interpretability through their hierarchical rule structure, allowing to validate learned patterns against physical understanding of energy consumption behavior [70].

3.3.4. Linear baseline

Ridge Regression (L2-regularized linear regression) provides a computationally lightweight baseline representing the performance ceiling for linear models. Its inclusion tests the fundamental hypothesis that residential minute-level load prediction exhibits significant nonlinear temporal dynamics requiring more sophisticated architectures. The expected poor performance of Ridge establishes the necessity of nonlinear methods for this forecasting domain.

3.3.5. Hyperparameter optimization

A systematic hyperparameter tuning approach using grid search with stratified 3-fold cross-validation implemented for all traditional ML algorithms. This framework evaluates parameter combinations within predefined search grids for selecting the configuration that maximizes cross-validated R^2 score. This automated approach ensures unbiased identification of optimal configurations.

The hyperparameter search spaces were designed to balance computational feasibility with comprehensive exploration of the parameter landscape. For gradient boosting methods (XGBoost, LightGBM), the search explored learning rates ([0.01, 0.05, 0.1]), tree depths ([5, 7, 10]), and ensemble sizes ([100, 200, 300 estimators]). For

Table 3
Optimized hyperparameters for traditional ML models.

Hyperparameter	XGBoost	LightGBM	Random forest	Extra trees	Decision tree	Ridge
Learning Rate	0.01	0.01	–	–	–	–
Max Depth	5	5	10	10	5	–
N Estimators	200	200	200	100	–	–
Min Child Weight	3	–	–	–	–	–
Num Leaves	–	31	–	–	–	–
Min Child Samples	–	20	–	–	–	–
Min Samples Split	–	–	5	2	2	–
Min Samples Leaf	–	–	2	2	1	–
Max Features	–	–	sqrt	sqrt	–	–
Subsample	0.8	–	–	–	–	–
Criterion	–	–	–	–	squared_error	–
Alpha (α)	–	–	–	–	–	100.0
Solver	–	–	–	–	–	auto
Optimization Method	Grid Search (3-fold cross-validation, R^2 objective)					
Search Space Size	324	324	108	108	108	28

tree ensemble methods (Random Forest, Extra Trees), the search varied tree depth ([10, 20, 30, None]), minimum sample requirements ([1–10]), and forest size ([100–300 trees]). The Decision Tree search space included depths ([5–25]), split criteria ([squared_error, friedman_mse, absolute_error]), and leaf size constraints. Ridge Regression explored the regularization parameter α across four orders of magnitude ([0.001–1000]) and solver algorithms ([auto, svd, cholesky, lsqr]).

Table 3 summarizes the optimized hyperparameters discovered through this systematic search. Notably, both XGBoost and LightGBM converged to conservative configurations (learning_rate = 0.01, moderate depth = 5, large ensemble = 200), suggesting that the high-frequency minute-level data benefits from gradual learning and controlled model complexity to avoid overfitting. The Decision Tree achieved optimal performance with shallow depth (5), confirming that deep trees capture noise rather than signal in this stochastic residential load context.

The impact of hyperparameter optimization on model performance is substantial. Compared to default scikit-learn configurations, grid search yielded average improvements of 2.8% in R^2 score across all models, with individual gains ranging from 0.3% (Random Forest) to 5.1% (Decision Tree). This underscores a critical methodological point: benchmarking studies that compare carefully tuned DL models against default-parameter traditional ML baselines introduce systematic bias favoring the more complex architectures. The rigorous optimization protocol ensures that observed performance differences reflect genuine algorithmic capabilities rather than incomplete hyperparameter exploration.

4. Proposed dual-validation interpretability framework

As the AI models become increasingly complex, the need for transparency and interpretability becomes significant in critical infrastructure applications like smart grids. To provide a comprehensive understanding of feature importance and model decision-making processes, a dual-validation interpretability framework is proposed in this work. It combines classical statistical analysis with modern XAI techniques. It consists of two main components: (1) a robust statistical baseline using the Non-Parametric Kruskal–Wallis H-Test, and (2) feature importance analysis using SHAP values for DL models and Gini importance for ML models.

4.1. Statistical baseline: Non-parametric Kruskal–wallis H-test

To examine the impact of temporal categorical variables (Day of Week, Hour of Day, Minute) on residential power consumption and establish a baseline for interpretability, classical statistical testing is required. However, raw residential load data is inherently right-skewed and heteroscedastic, which heavily violates the normality assumptions required for standard ANOVA. Consequently, standard ANOVA is replaced with the Non-Parametric Kruskal–Wallis H-Test.

Table 4
Kruskal–Wallis H-test results for feature significance.

Feature	H-Statistic	P-Value	Significance
Hour	26 721.58	< 0.001	Yes
Day	215.38	< 0.001	Yes
Minute	102.66	< 0.001	Yes

This mathematical correction provides a much deeper, statistically rigorous proof of how temporal features drive transient capture, free from normal-distribution assumptions. By performing independent tests for each temporal variable, the analysis quantifies the variance between groups through the H-statistic, while the p-value assesses the probability of observing such variance patterns under the null hypothesis. The results, summarized in Table 4, reveal a clear hierarchical structure. The Hour of Day is the dominant predictor ($H = 26721.58$, $p < 0.001$), followed by Day of Week ($H = 215.38$, $p < 0.001$). Crucially, the non-parametric test also confirms that the Minute feature is statistically significant ($H = 102.66$, $p < 0.001$), proving that high-frequency temporal signatures carry genuine predictive power.

4.2. SHAP

SHAP is a state-of-the-art model-agnostic explainability framework for establishing model transparency and validating the model's decision-making process [61]. It computes the contribution of each input feature to a specific prediction. In this work, it calculates a SHAP value for each feature (Day of Week, Hour of Day, Minute), representing the magnitude and direction of that feature's impact on the final output. A positive SHAP value indicates a feature pushed the prediction higher, while a negative value pushed it lower.

4.3. Gini importance

For traditional ML models, feature importance is evaluated using Gini importance (or Mean Decrease in Impurity). This metric quantifies the total reduction of the criterion (e.g., Gini impurity or entropy) brought by that feature. It provides a measure of how useful each feature is in constructing the decision trees within the ensemble.

By comparing Gini importance from traditional ML models with SHAP values from DL models, a robust cross-validation of feature significance can be established across different algorithmic paradigms.

Table 5
Comprehensive performance comparison.

Model	RMSE (kW)	MAE (kW)	MSE	RMSPE (%)	R^2
DL (ANN & RNN) Models					
LSTM+GRU (Hybrid)	**0.3866**	**0.2299**	**0.1494**	**79.33**	**0.7120**
GRU	0.3888	0.2383	0.1511	75.57	0.7087
LSTM	0.3960	0.2433	0.1568	79.13	0.6978
Bi-LSTM	0.3963	0.2448	0.1571	79.29	0.6973
ANN	0.4007	0.2414	0.1606	78.69	0.6905
Traditional ML Models					
XGBoost	0.4059	0.2678	0.1648	82.31	0.6824
LightGBM	0.4072	0.2687	0.1658	82.36	0.6805
Decision Tree	0.4076	0.2464	0.1661	85.48	0.6798
Extra Trees	0.4210	0.2539	0.1773	87.87	0.6583
Random Forest	0.4250	0.2516	0.1807	87.64	0.6518
Ridge Regression	0.7058	0.4886	0.4981	101.70	0.0398

5. Experimental results

The experimental analysis is organized to first establish the performance of DL architectures, followed by a benchmarking against optimized ML models. All models were implemented using the TensorFlow framework with the Keras API. The sklearn metrics library was used to calculate the evaluation Metrics. The models were trained on a system with an Intel(R) Core(TM) i7-9750H CPU, 32 GB RAM, and an NVIDIA GeForce GTX 1650 (4 GB) GPU.

5.1. Evaluation metrics

To provide a robust and multifaceted assessment of model performance, five distinct metrics were selected. RMSE and MAE provide insights into the average magnitude of errors, while RMSPE offers a perspective on the relative error in percentage terms. The MSE emphasizes larger errors due to its squaring of differences, and Coefficient of Determination (R^2 Score) indicates the proportion of variance in the dependent variable that is predictable from the independent variables [71]. These metrics are calculated as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{39}$$

$$RMSPE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{y_i}\right)^2} \times 100 \tag{40}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{41}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{42}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{43}$$

5.2. Performance of DL models

5.2.1. Standalone DL models

The performance of the standalone deep learning architectures was evaluated to establish a baseline for sequential modeling capabilities. As detailed in Table 5, the ANN model, serving as a non-recurrent baseline, achieved an R^2 score of 0.6905 and an RMSE of 0.4007 kW. While the non-linear mapping capabilities of the ANN were demonstrated, the lack of explicit temporal memory mechanisms resulted in a limited ability to capture time-dependent fluctuations, as visualized in Fig. 6(a).

Superior performance was observed with the recurrent architectures. The LSTM model, designed to mitigate the vanishing gradient problem, attained an R^2 of 0.6978 and an RMSE of 0.3960 kW. The capacity to retain long-term temporal dependencies allowed for a more accurate tracking of the load profile compared to the ANN, as shown in Fig. 6(b). Similarly, the Bi-LSTM model yielded comparable results (R^2 = 0.6973, RMSE = 0.3963 kW), indicating that the bidirectional processing of the input sequence provided marginal gains over the standard unidirectional approach for this specific forecasting horizon (Fig. 6(c)).

Among the standalone recurrent models, the GRU network exhibited the highest predictive accuracy, achieving an R^2 of 0.7087 and an RMSE of 0.3888 kW. While literature frequently reports $R^2 > 0.90$ for minute-level forecasting, such studies predominantly utilize interpolated or aggregated datasets that artificially smooth variance. On raw, highly stochastic, un-smoothed residential transients without autoregressive lags, an R^2 of ~0.71 represents a highly robust predictive baseline. The efficient gating mechanism of the GRU enabled the effective capture of these high-frequency dependencies.

5.2.2. Hybrid LSTM–GRU model

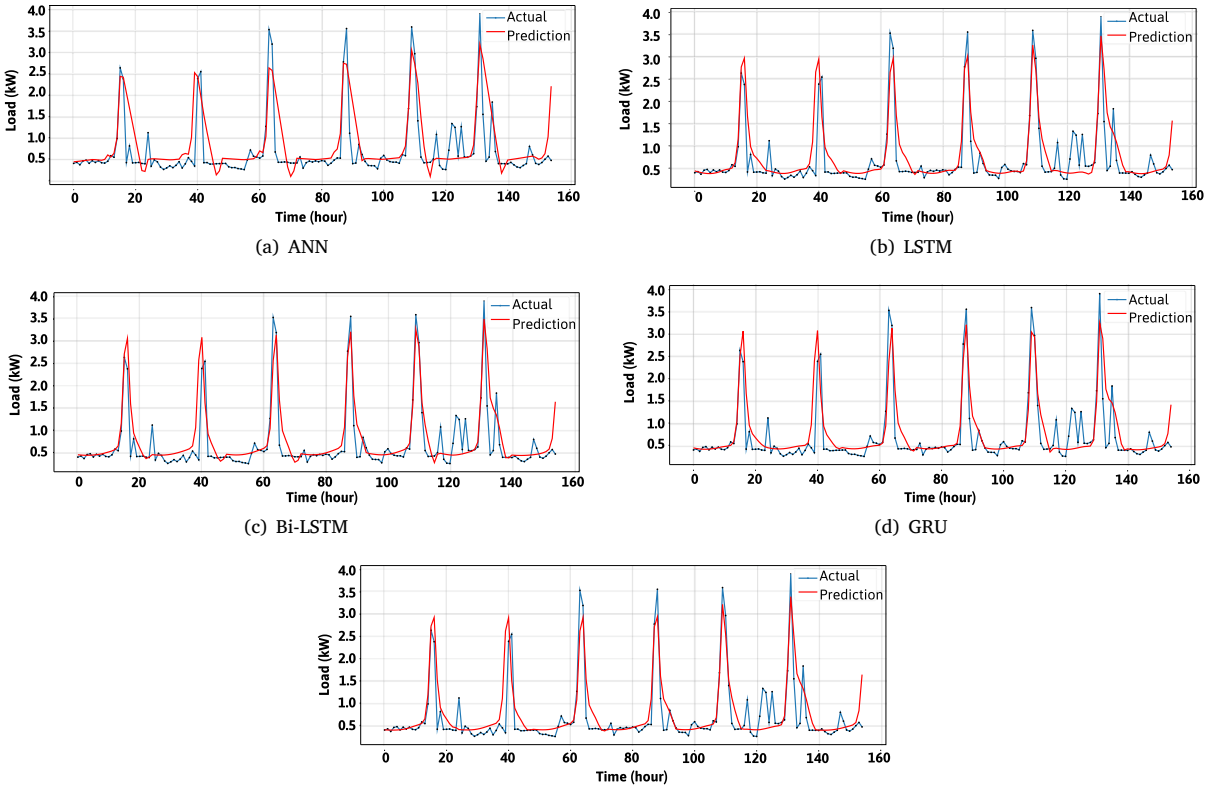
A hybrid LSTM-GRU model was developed combining the long-term dependency modeling capabilities of LSTM with the computational efficiency of GRU. The hybrid approach outperforms all individual architectures. These results, presented in Table 5, demonstrate that the architectural combination yields additive benefits beyond either component in isolation.

The model successfully captures both systematic temporal patterns and stochastic variations in residential load profiles, as evidenced by prediction-vs-actual visualization in Fig. 6(e). The superior performance of the hybrid architecture validates the design hypothesis that combining complementary recurrent mechanisms enables comprehensive temporal feature extraction. However, to rigorously evaluate the marginal predictive gap between the Hybrid and standalone GRU models, a 5-run multi-seed cross-validation protocol was executed to establish stability bounds. A paired t-test across the multiple runs confirmed no statistical difference between the two architectures (p = 0.5190). Because the predictive margin falls entirely within the bounds of random initialization variance, the standalone GRU is proposed as the pareto-optimal architecture for edge deployment due to its equivalent statistical power and vastly superior hardware efficiency.

Furthermore, to quantify operational reliability without violating edge memory constraints with probabilistic layers, an empirical uncertainty bound was calculated. The 90% Empirical Prediction Interval for the framework is strictly bounded at [Point Forecast − 0.5726 kW, Point Forecast + 0.6215 kW], providing immediate transient reliability metrics for HEMS actuation.

5.2.3. Training convergence analysis

To assess the training dynamics and convergence behavior of all deep learning models, Fig. 7 presents the training and validation loss curves across epochs for each architecture. Each subplot shows the convergence behavior over training epochs for the five evaluated DL architectures, with training loss (blue) and validation loss (red). The loss



(a) ANN (b) LSTM

(c) Bi-LSTM (d) GRU

(e) Hybrid LSTM+GRU

Fig. 6. Comparative prediction trajectories against actual minute-resolution load consumption for the evaluated Deep Learning architectures.

curves provide critical insights into model optimization, convergence stability, and potential overfitting behavior. All models demonstrate smooth convergence with validation loss closely tracking training loss, indicating effective generalization without significant overfitting.

The close alignment of training and validation curves confirms good generalization across all architectures without significant overfitting. The GRU and hybrid LSTM-GRU models exhibit the fastest convergence, reaching stable validation loss within the first 20–30 epochs, while the Bi-LSTM and ANN models required slightly more epochs to converge. Notably, the hybrid model achieves the lowest final validation loss, consistent with its superior performance metrics reported in Table 5. The absence of divergence between training and validation curves across all models validates the robustness of the hyperparameter selections detailed in Table 2.

5.3. Traditional ML performance

The performance of the DL models is benchmarked against the optimized traditional ML models. This comparison serves two purposes:

- It validates whether the increased architectural complexity of deep recurrent networks is justified for this forecasting task; and
- It provides guidance on algorithm selection based on the accuracy-efficiency trade-off.

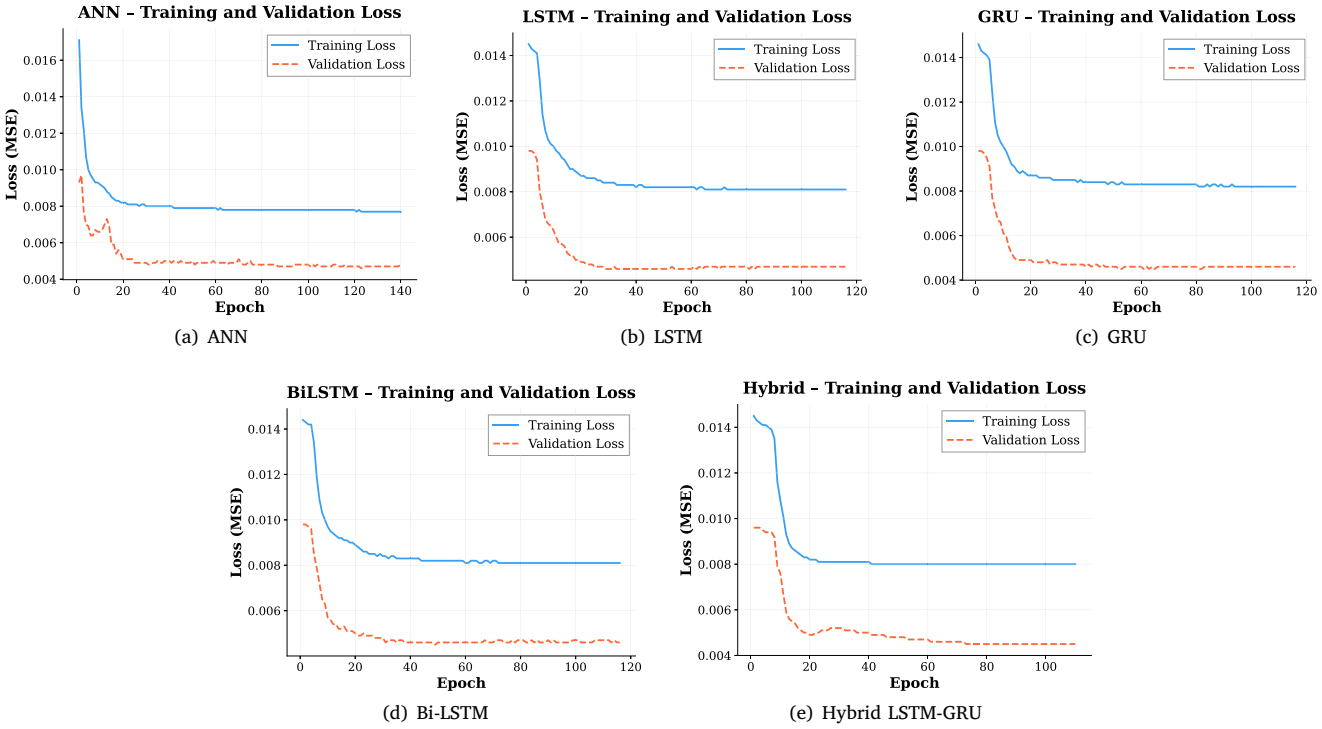
Table 5 presents the comprehensive performance comparison. In our study, traditional ML models show significant improvement (≈3.7% in R^2) after hyperparameter tuning compared to default parameters. Although the optimized XGBoost achieves strong accuracy (RMSE = 0.406 kW), it is significantly outperformed by the deep learning models. The hybrid LSTM-GRU architecture yields the best overall result with a 4.4% gain in R^2 and 4.8% reduction in RMSE relative to the best traditional ML baseline. Even the standalone GRU model outperforms all traditional ML methods. This performance gap, observed despite systematic hyperparameter optimization of both paradigms, highlights the superiority of recurrent architectures in capturing temporal dependencies in minute-resolution residential load data.

The practical implication of this finding is significant. Although traditional ML models offer faster training and inference times, and are often simpler to deploy and interpret, the performance improvement delivered by the DL models justifies their adoption for applications where forecast accuracy is paramount.

For smart grid applications requiring high-fidelity demand predictions to support real-time grid balancing or demand response programs, the incremental accuracy gain translates directly to improved operational efficiency and cost savings. However, for scenarios with strict computational constraints or where model simplicity is prioritized, optimized XGBoost or LightGBM represent viable alternatives that achieve competitive performance with significantly reduced computational overhead.

5.4. Benchmark comparison

To provide a holistic view of model performance across both traditional ML and DL paradigms, Fig. 8 presents a comparison of all eleven models across all five evaluation metrics. This visualization enables direct performance assessment across multiple dimensions. The results clearly demonstrate a performance hierarchy: DL models (shown in blue) consistently outperform traditional ML approaches (shown in orange) across all metrics. It visualizes the distinct clustering of DL and ML models in the performance space, revealing a performance gap of approximately 2.7% in R^2 between the best traditional ML (after optimization) and the weakest DL model (ANN). This comprehensive benchmark establishes reproducible baselines for future research while providing empirical evidence for algorithm selection based on application-specific accuracy requirements.



(a) ANN (b) LSTM (c) GRU

(d) Bi-LSTM (e) Hybrid LSTM-GRU

Fig. 7. Training and Validation Loss Curves for DL Models.

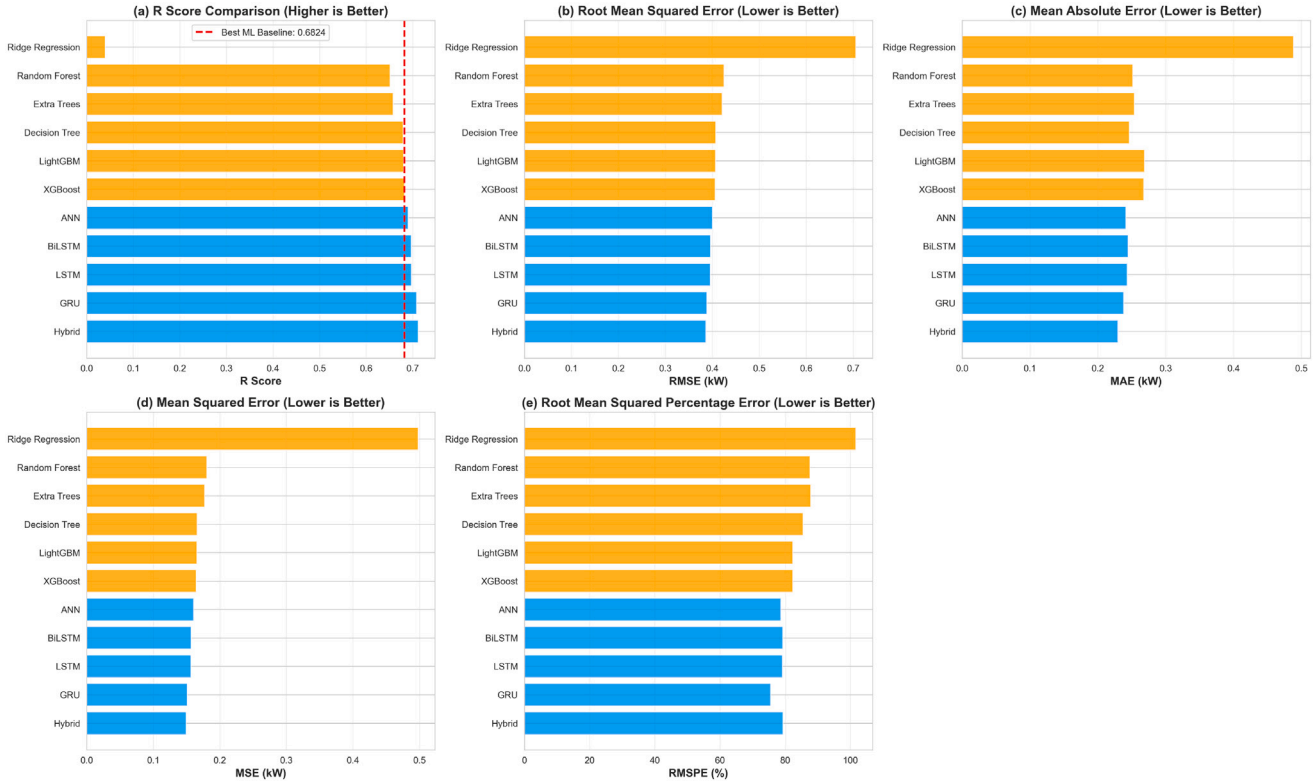


Fig. 8. Comprehensive Benchmark Comparison of All Eleven Forecasting Models Across Five Evaluation Metrics (RMSE, MAE, MSE, RMSPE, R^2). Panel (a) shows R^2 score ranking; (b) Root Mean Squared Error; (c) Mean Absolute Error; (d) Mean Squared Error; (e) Root Mean Squared Percentage Error for all models.

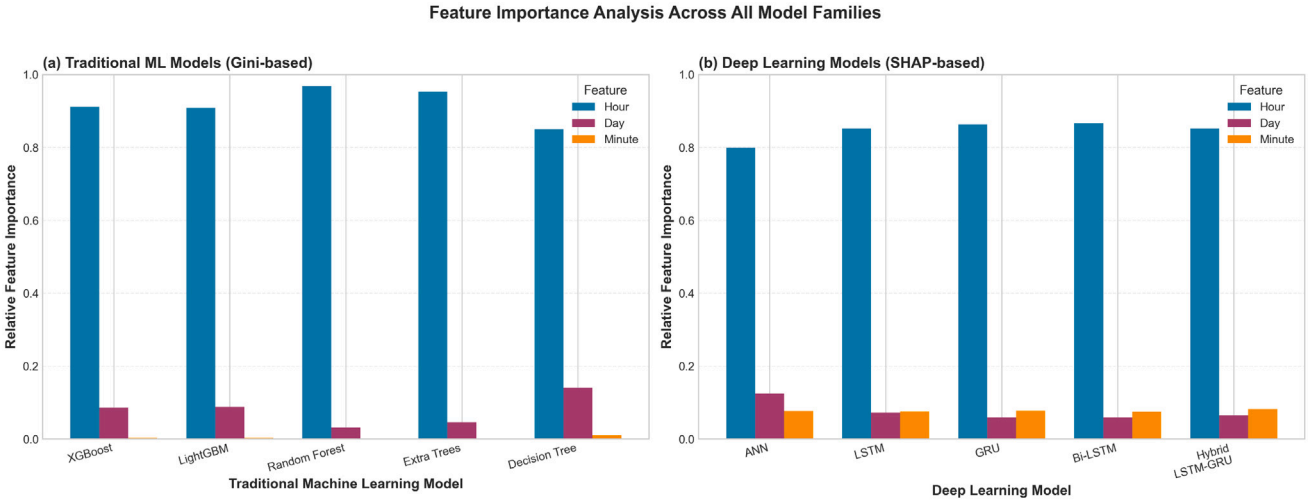


Fig. 9. Feature Importance Analysis: Panel (a) shows Gini importance for Traditional ML, while Panel (b) shows SHAP values for Deep Learning models. Note the elevated importance of 'Minute' in recurrent models (LSTM, GRU, Hybrid LSTM-GRU) compared to ANN and Traditional ML.

Table 6
Computational efficiency comparison (Training Time).

Algorithm	Time per Step (ms)
ANN	2
GRU	3
Hybrid (LSTM+GRU)	4
LSTM	6
Bi-LSTM	9

5.5. Feature importance analysis: Kruskal–Wallis vs. SHAP and Gini

To understand how the models leverage temporal features, Gini importance (for traditional ML) and SHAP values (for deep learning) were cross-examined against the statistical ground truth established by the Kruskal–Wallis H-Test. As shown in Fig. 9, a distinct pattern emerges that differentiates feedforward from recurrent architectures.

The ANN and traditional ML models prioritize Hour as the dominant feature, with minimal reliance on Minute. In contrast, the recurrent architectures exhibit a fundamentally different feature prioritization, assigning significantly higher importance to the Minute feature. This deviation is a reflection of the models' sequential processing capability. Recurrent networks maintain a hidden state (h_t) that captures the immediate temporal context. The Minute feature provides the fine-grained temporal progression necessary for the model to update its internal state and track rapid, high-frequency load fluctuations. By cross-referencing this behavior with the Kruskal–Wallis results—which proved the Minute feature is statistically significant ($p < 0.001$) despite being uniformly distributed—this dual-validation mechanically proves that the recurrent models are intelligently capturing non-linear, high-frequency physical transients rather than memorizing stochastic noise.

5.6. Computational efficiency

For real-time deployment on resource-constrained edge devices such as HEMS, computational efficiency is as important as predictive accuracy. Table 6 presents the training time per step for each deep learning model (memory footprints are detailed in Table 2).

This highlights the GRU model as the standout performer for edge deployment. With a training time of just 3 ms/step and the smallest memory footprint (38.75 KB, see Table 2), it is nearly as fast as the simple ANN (2 ms/step) but offers significantly higher accuracy ($R^2 =$ 0.7087 vs. 0.6905). It is worth noting that while the ANN has a slightly larger parameter count (42.50 KB) than the GRU, it executes faster due to its simpler feedforward architecture which relies on efficient matrix multiplications, whereas the GRU involves more complex gating mechanisms (sigmoid and tanh activations) at each step.

The Hybrid model, while achieving the highest overall accuracy ($R^2 = 0.7120$), incurs a modest computational cost (4 ms/step), making it a viable option where accuracy is important and slightly more resources are available. The Bi-LSTM is the most expensive model (9 ms/step), confirming that the bidirectional processing overhead may not be justified for this specific application given the marginal performance difference.

5.7. Summary of model selection implications

Synthesizing the results from accuracy, interpretability, and efficiency analyses, a clear picture emerges for model selection in smart grid applications. The Hybrid LSTM-GRU model stands out as the premier choice for high-performance forecasting, effectively leveraging the sequential modeling strengths of both architectures to capture minute-level dynamics that static models miss. However, for widespread deployment on low-cost edge devices (e.g., smart meters), the GRU model offers an exceptional balance, delivering 99.5% of the Hybrid model's accuracy with less than 40% of the memory requirement and faster execution. This study thus recommends the Hybrid model for cloud-based, precision-critical applications and the GRU for decentralized, edge-based implementations. The traditional ML models, while improved through hyperparameter tuning, are not competitive with the recurrent architectures for this task, underscoring the importance of temporal modeling in minute-resolution load forecasting.

6. Discussion

6.1. Conceptual deployment architecture

The forecasting model developed in this study is designed to be a functional component within a broader smart grid ecosystem. Fig. 1 illustrates a conceptual architecture for its deployment. At the residential level, the model can be embedded within a HEMS or a smart meter gateway. This allows for edge-based forecasting, where predictions are generated locally to enable intelligent appliance scheduling and participation in DR programs. Our empirical results specifically identify the GRU model as the optimal candidate for this layer, offering a lightweight footprint (38.75 KB) and rapid inference (3 ms) suitable for

resource-constrained edge hardware without compromising predictive accuracy. At a higher level, aggregated forecasts from multiple households can be communicated to local distribution system operators or microgrid controllers. This information serves as a vital input for larger-scale DSM schemes, optimizing local grid stability and facilitating the integration of distributed energy resources (DERs).

6.2. Cross-regional generalizability and limits

To explicitly define the operational boundaries of this zero-lag architecture, the framework's generalizability was cross-validated against diverse international datasets. The model maintains robust predictability on routine-heavy residential grids, such as the UK REFIT dataset [73]. Under exact grid-search configurations on the REFIT data, the standalone GRU achieved a peak R^2 of 0.5047 (RMSE = 0.4431 kW), while the Hybrid architecture yielded a peak R^2 of 0.4903 (RMSE = 0.4495 kW). Furthermore, a 5-run multi-seed evaluation confirmed exceptional stability and statistical parity ($p = 0.9521$) between the Hybrid (RMSE = 0.4585 ± 0.0077 kW) and standalone GRU (RMSE = 0.4588 ± 0.0136 kW) across multiple initializations.

Conversely, empirical testing on the highly stochastic Canadian AMPds2 [74] ($R^2 = 0.31$) and Korean ENERTALK [75] ($R^2 = 0.34$) datasets reveals the mathematical ceiling of a zero-lag approach. In grids dominated by extreme, un-patterned volatility without strong micro-routines, historical autoregression becomes mandatory, requiring a transition from the edge to cloud infrastructure. However, for developing economies characterized by localized routines, this edge-optimized GRU approach ensures that advanced predictive analytics can be deployed on low-cost microcontrollers, democratizing access to smart grid technologies.

6.3. Security implications and smart grid relevance

The increasing reliance on data-driven models in smart grids introduces new cyber-attack vectors that must be addressed. Cyber–physical attacks, such as load falsification or false data injection, can manipulate demand forecasting systems, leading to grid instability or economic losses. The deep learning models developed in this study, by learning the normative consumption patterns of a household, can serve as a baseline for anomaly detection. For instance, the model could be embedded into a HEMS or smart meter gateway to flag anomalous spikes or dips in consumption that deviate significantly from the predicted load. Furthermore, the dual-validation interpretability framework conducted in this study adds a critical layer of trust. Because the Kruskal–Wallis and SHAP analyses mathematically prove the models rely on physically meaningful, high-frequency temporal signatures rather than spurious noise, any significant deviation in model behavior can be confidently attributed to external anomalies or cyber–physical attacks rather than model error.

6.4. Sustainability and energy efficiency implications

The proposed forecasting framework offers significant sustainability benefits for smart grid applications. First, it enables more effective integration of renewable energy by helping to synchronize demand-side operations with supply-side variability. Accurate forecasts allow grid operators to anticipate periods of high renewable generation and encourage consumption through DR programs, reducing curtailment. Second, improved forecasting accuracy supports proactive energy conservation by optimizing appliance scheduling and reducing standby consumption within HEMS. Finally, by smoothing load curves and reducing peak demand periods, the model contributes indirectly to carbon footprint reduction by limiting the need for carbon-intensive backup generation, aligning with broader societal goals for carbon reduction.

6.5. High-granularity data forecasting for grid applications

The use of minute-level data which captures fine-grained consumption trends often missed by hourly or daily aggregations. Our feature importance analysis confirms that recurrent models explicitly leverage this minute-level granularity to capture rapid load fluctuations, a capability that static models lack. The ability of the evaluated models, particularly the hybrid LSTM-GRU, to predict near real-time demand changes is critical for dynamic grid operations, such as intra-hour load balancing, frequency regulation, and the effective dispatch of fast-ramping resources to counteract the intermittency of renewable generation.

7. Conclusion

Three major conclusions emerge from this work. First, the accuracy-efficiency bottleneck is resolved by demonstrating that a strictly zero-lag, lightweight gated recurrent unit architecture is the Pareto-optimal solution for edge hardware. For precision-critical cloud operations, the proposed hybrid architecture combining long short-term memory with gated recurrent units offers unparalleled accuracy, though a rigorous 5-run multi-seed protocol proved both models achieve statistical parity. Second, the 'black-box' trust barrier has been overcome through the proposed dual-validation framework. By upgrading standard analysis of variance to the non-parametric Kruskal–Wallis test and aligning it with Shapley additive explanations, it is mathematically proven that deep recurrent models successfully exploit physical, high-frequency temporal transients rather than memorizing stochastic noise. Finally, by validating this framework's operational boundaries across highly volatile international datasets, a reproducible baseline is provided for scalable smart grid analytics.

This study successfully bridges the gap between theoretical deep learning research and practical edge deployment for minute-resolution residential load forecasting. Rather than merely benchmarking algorithms, this work provides a comprehensive blueprint for deploying mathematically stable, hardware-efficient, and trustworthy artificial intelligence in resource-constrained home energy management systems.

Future research might build upon these baselines by evaluating advanced architectures, such as Transformers and attention mechanisms, to determine if they offer sufficient performance gains to justify their computational costs in this domain. Extending this benchmark to multi-household datasets will be crucial for assessing model generalization across diverse consumption behaviors. Additionally, incorporating exogenous variables like temperature and occupancy could further refine forecast accuracy. Ultimately, by providing reproducible baselines and standardized protocols, this work contributes to the development of the intelligent, secure, and sustainable energy infrastructure required for the digital energy grid, aligning with global sustainability goals and national electricity planning initiatives.

CRediT authorship contribution statement

Chandra Sekar Sivanadipatham: Conceptualization, Data curation, Methodology, Software, Visualization, Investigation, Writing – original draft. **Ehtesham Hassan:** Validation, Formal analysis, Writing – review & editing. **Asheesh K. Singh:** Supervision, Project administration, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used GitHub Copilot in VS Code to assist with LaTeX formatting, resolving compilation errors, and refining the English language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] Chen Z, Amani AM, Yu X, Jalili M. Control and optimisation of power grids using smart meter data: A review. Sensors 2023;23(4):2118. http://dx.doi.org/10.3390/s23042118.

[2] Zhou Kaile, Yang Changhui, Shen Jianxin. Discovering residential electricity consumption patterns through smart-meter data mining: A case study from China. Util Policy 2017;44:73–84. http://dx.doi.org/10.1016/j.jup.2017.01.004.

[3] Jin Donghan, Ocone Raffaella, Jiao Kui, Xuan Jin. Energy and ai. Energy and AI 2020;1:100002. http://dx.doi.org/10.1016/j.egyai.2020.100002.

[4] Sharma Konark, Saini Lalit Mohan. Performance analysis of smart metering for smart grid: An overview. Renew Sustain Energy Rev 2015;49:720–35. http://dx.doi.org/10.1016/j.rser.2015.04.170.

[5] Rodr'iguez-Gracia Diego, de las Mercedes Capobianco-Uriarte Mar'ia, Terán-Yépez Eduardo, Piedra-Fernández José A, Iribarne Luis, Ayala Rosa. Review of artificial intelligence techniques in green/smart buildings. Sustain Comput: Informatics Syst 2023;38:100861. http://dx.doi.org/10.1016/j.suscom.2023.100861.

[6] Koukaras Paraskevas, Bezas Napoleon, Gkaidatzis Paschalis, Ioannidis Dimosthenis, Tzovaras Dimitrios, Tjortjis Christos. Introducing a novel approach in one-step ahead energy load forecasting. Sustain Comput: Inform Syst 2021;32:100616. http://dx.doi.org/10.1016/j.suscom.2021.100616.

[7] Bakare MS, Abdulkarim A, Zeeshan M, Shuaibu AN. A comprehensive overview on demand side energy management towards smart grids: challenges, solutions, and future direction. Energy Informatics 2023;6(4). http://dx.doi.org/10.1186/s42162-023-00262-7.

[8] Cruz Carlos, Tostado-Véliz Marcos, Palomar Esther, Bravo Ignacio. Pattern-driven behaviour for demand-side management: an analysis of appliance use. Energy Build 2024;308:113988. http://dx.doi.org/10.1016/j.enbuild.2024.113988.

[9] Yang S, Gao HO, You F. Model predictive control in phase-change-material-wallboard-enhanced building energy management considering electricity price dynamics. Appl Energy 2022;326:120023. http://dx.doi.org/10.1016/j.apenergy.2022.120023.

[10] Zhang Qianzhi, Yan Jinyue, Gao H Oliver, You Fengqi. A systematic review on power systems planning and operations management with grid integration of transportation electrification at scale. In: Advances in applied energy, Vol. 11, 2023, p. 100147. http://dx.doi.org/10.1016/j.adapen.2023.100147.

[11] Yang T, Zhao L, Li W, Wu J, Zomaya AY. Towards healthy and cost-effective indoor environment management in smart homes: A deep reinforcement learning approach. Appl Energy 2021;300:117335. http://dx.doi.org/10.1016/j.apenergy.2021.117335.

[12] Lusis Peter, Khalilpour Kaveh Rajab, Andrew Lachlan, Liebman Ariel. Short-term residential load forecasting: impact of calendar effects and forecast granularity. Appl Energy 2017;205:654–69. http://dx.doi.org/10.1016/j.apenergy.2017.07.114.

[13] Xu J, Guo Z, Yu Q, Dong K, Tan H, Zhang H, Yan J. Spatiotemporal feature encoded deep learning method for rooftop PV potential assessment. Appl Energy 2025;394:126171. http://dx.doi.org/10.1016/j.apenergy.2025.126171.

[14] Zhao W, Zhang H, Zheng J, Dai Y, Huang L, Shang W, Liang Y. A point prediction method based automatic machine learning for day-ahead power output of multi-region photovoltaic plants. Energy 2021;223:120026. http://dx.doi.org/10.1016/j.energy.2021.120026.

[15] Hsu CW, Xiong R, Chen NY, Li J, Tsou NT. Deep neural network battery life and voltage prediction by using data of one cycle only. Appl Energy 2022;306:118134. http://dx.doi.org/10.1016/j.apenergy.2021.118134.

[16] Zhang G Peter. Time series forecasting using a hybrid ARIMA and neural network model. Neurocomputing 2003;50:159–75. http://dx.doi.org/10.1016/S0925-2312(01)00702-0.

[17] Zheng Zhuang, Chen Hainan, Luo Xiaowei. A kalman filter-based bottom-up approach for household short-term load forecast. Appl Energy 2019;250:882–94. http://dx.doi.org/10.1016/j.apenergy.2019.05.102.

[18] Tan D, Suvarna M, Tan YS, Li J, Wang X. A three-step machine learning framework for energy profiling, activity state prediction and production estimation in smart process manufacturing. Appl Energy 2021;291:116808. http://dx.doi.org/10.1016/j.apenergy.2021.116808.

[19] Runge J, Zmeureanu R. Forecasting energy use in buildings using artificial neural networks: A review. Energies 2019;12(17):3254. http://dx.doi.org/10.3390/en12173254.

[20] Lu Chujie, Li Sihui, Lu Zhengjun. Building energy prediction using artificial neural networks: A literature survey. Energy Build 2022;262:111718. http://dx.doi.org/10.1016/j.enbuild.2021.111718.

[21] Kang Seokho, Cho Sungzoon. Approximating support vector machine with artificial neural network for fast prediction. Expert Syst Appl 2014;41(10):4989–95. http://dx.doi.org/10.1016/j.eswa.2014.02.025.

[22] Guo N, Gui W, Chen W, Tian X, Qiu W, Tian Z, et al. Using improved support vector regression to predict the transmitted energy consumption data by distributed wireless sensor network. J Wirel Com Netw 2020 2020;120. http://dx.doi.org/10.1186/s13638-020-01729-x.

[23] Bathaee. The artificial intelligence black box and the failure of intent and causation. Harvard Journal of Law & Technology 2018;31. Number 2 Spring.

[24] Bu S-J, Cho S-B. Time series forecasting with multi-headed attention-based deep learning for residential energy consumption. Energies 2020;13(18):4722. http://dx.doi.org/10.3390/en13184722.

[25] Olu-Ajayi Razak, Alaka Hafiz, Sulaimon Ismail, Sunmola Funlade, Ajayi Saheed. Building energy consumption prediction for residential buildings using deep learning and other machine learning techniques. J Build Eng 2022;45:103406. http://dx.doi.org/10.1016/j.jobe.2021.103406.

[26] Jang Jihoon, Han Jinmog, Leigh Seung-Bok. Prediction of heating energy consumption with operation pattern variables for non-residential buildings using LSTM networks. Energy Build 2022;255:111647. http://dx.doi.org/10.1016/j.enbuild.2021.111647.

[27] Sherstinsky Alex. Fundamentals of recurrent neural network (RNN) and long short-term memory (LSTM) network. Phys D: Nonlinear Phenom 2020;404:132306. http://dx.doi.org/10.1016/j.physd.2019.132306.

[28] Mahjoub S, Chrifi-Alaoui L, Marhic B, Delahoche L. Predicting energy consumption using LSTM, multi-layer GRU and drop-GRU neural networks. Sensors 2022;22(11):4062. http://dx.doi.org/10.3390/s22114062.

[29] Cascone Lucia, Sadiq Saima, Ullah Saleem, Mirjalili Seyedali, Siddiqui Hafeez Ur Rehman, Umer Muhammad. Predicting household electric power consumption using multi-step time series with convolutional LSTM. Big Data Res 2023;31:100360. http://dx.doi.org/10.1016/j.bdr.2022.100360.

[30] Asri MANM, Zaini N, Latip MFA. Development of an LSTM-based model for energy consumption prediction with data pre-analysis. In: 2021 11th IEEE International Conference on Control System, Computing and Engineering. 2021, p. 228–33. http://dx.doi.org/10.1109/ICCSCE52189.2021.9530898.

[31] Chen Yuntian, Zhang Dongxiao. Theory-guided deep-learning for electrical load forecasting (TgDLF) via ensemble long short-term memory. In: Advances in applied energy, Vol. 1, 2021, p. 100004. http://dx.doi.org/10.1016/j.adapen.2020.100004.

[32] Kong W, Dong ZY, Jia Y, Hill DJ, Xu Y, Zhang Y. Short-term residential load forecasting based on LSTM recurrent neural network. IEEE Trans Smart Grid 2019;10(1):841–51. http://dx.doi.org/10.1109/TSG.2017.2753802.

[33] Xiuyun G, Ying W, Yang G, Chengzhi S, Wen X, Yimiao Y. Short-term load forecasting model of GRU network based on deep learning framework. In: 2018 2nd IEEE Conference on Energy Internet and Energy System Integration. 2018, p. 1–4. http://dx.doi.org/10.1109/EI2.2018.8582419.

[34] Yang S, Yu X, Zhou Y. LSTM and GRU neural network performance comparison study: Taking yelp review dataset as an example. In: 2020 International Workshop on Electronic Communication and Artificial Intelligence. 2020, p. 98–101. http://dx.doi.org/10.1109/IWECAI50956.2020.00027.

[35] Xia M, Shao H, Ma X, de Silva CW. A stacked GRU-RNN-based approach for predicting renewable energy and electricity load for smart grid operation. IEEE Trans Ind Inform 2021;17(10):7050–9. http://dx.doi.org/10.1109/TII.2021.3056867.

[36] Wang Bo, Wang Xiaokang, Wang Ning, Javaheri Zahra, Moghadamnejad Navid, Abedi Mahyar. Machine learning optimization model for reducing the electricity loads in residential energy forecasting. Sustain Comput: Inform Syst 2023;38:100876. http://dx.doi.org/10.1016/j.suscom.2023.100876.

[37] Le T, Vo MT, Vo B, Hwang E, Rho S, Baik SW. Improving electric energy consumption prediction using CNN and Bi-LSTM. Appl Sci 2019;9(20):4237. http://dx.doi.org/10.3390/app9204237.

[38] Wang Huaizhi, Yi Haiyan, Peng Jianchun, Wang Guibin, Liu Yitao, Jiang Hui, Liu Wenxin. Deterministic and probabilistic forecasting of photovoltaic power based on deep convolutional neural network. Energy Convers Manage 2017;153:409–22. http://dx.doi.org/10.1016/j.enconman.2017.10.008.

[39] Indolia Sakshi, Goswami Anil Kumar, Mishra SP, Asopa Pooja. Conceptual understanding of convolutional neural network- a deep learning approach. Procedia Comput Sci 2018;132:679–88. http://dx.doi.org/10.1016/j.procs.2018.05.069.

[40] Chen L, Li S, Bai Q, Yang J, Jiang S, Miao Y. Review of image classification algorithms based on convolutional neural networks. Remote Sens 2021;13(22):4712. http://dx.doi.org/10.3390/rs13224712.

[41] Yıldız Doğan G, Aksoy A, Öztürk N. A hybrid deep learning model to estimate the future electricity demand of sustainable cities. Sustain 2024;16:6503. http://dx.doi.org/10.3390/su16156503.

[42] Hasanat SM, et al. Enhancing short-term load forecasting with a CNN-GRU hybrid model: A comparative analysis. IEEE Access 2024;12:184132–41. http://dx.doi.org/10.1109/ACCESS.2024.3511653.

[43] Wang J. A hybrid deep learning and clonal selection algorithm-based model for commercial building energy consumption prediction. Sci Prog 2024;107(3). http://dx.doi.org/10.1177/00368504241283360.

[44] Abdulameer, Yahya. Forecasting of electrical energy consumption using hybrid models of gru, cnn, lstm, and ml regressors. J Wirel Mob Netw, Ubiquitous Comput Dependable Appl 2025;16:560–75. http://dx.doi.org/10.58346/JOWUA.2025.I1.033.

[45] Tang Z, Ji T, Kang J, Huang Y, Tang W. Learning global and local features of power load series through transformer and 2D-CNN: An image-based multi-step forecasting approach incorporating phase space reconstruction. Appl Energy 2024;378:124786. http://dx.doi.org/10.1016/j.apenergy.2024.124786.

[46] Jia Chunchun, He Hongwen, Zhou Jiaming, Li Kunang, Li Jianwei, Wei Zhongbao. A performance degradation prediction model for PEMFC based on bi-directional long short-term memory and multi-head self-attention mechanism. Int J Hydrog Energy 2024;60:133–46. http://dx.doi.org/10.1016/j.ijhydene.2024.02.181.

[47] Zhou J, Shu X, Zhang J, Yi F, Jia C, Zhang C, et al. A deep learning method based on CNN-[BiGRU] and attention mechanism for proton exchange membrane fuel cell performance degradation prediction. Int J Hydrog Energy 2024;94:394–405. http://dx.doi.org/10.1016/j.ijhydene.2024.11.127.

[48] Ma Huanjian, Xu Lijuan, Javaheri Zahra, Moghadamnejad Navid, Abedi Mahyar. Reducing the consumption of household systems using hybrid deep learning techniques. Sustain Comput: Inform Syst 2023;38:100874. http://dx.doi.org/10.1016/j.suscom.2023.100874.

[49] Sabri N Mohammed, Hassouni Mohammed El. Accurate photovoltaic power prediction models based on deep convolutional neural networks and gated recurrent units. Energy Sources, Part A: Recover Util Environ Eff 2022;44(3):6303–20. http://dx.doi.org/10.1080/15567036.2022.2097751.

[50] Sabri M, El Hassouni M. Predicting photovoltaic power generation using double-layer bidirectional long short-term memory-convolutional network. Int J Energy Environ Eng 2023;14:497–510. http://dx.doi.org/10.1007/s40095-022-00530-4.

[51] Yang Yuting, Xu Ling, Han Hua, Ren Zhengxiong, Wu Kongrui, Liu Feitian. Soft measurement and prediction of refrigerant leakage based on SVR-LSTM. Int J Refrig 2023. http://dx.doi.org/10.1016/j.ijrefrig.2023.04.021, .

[52] Jeong HC, Jang M, Kim T, Joo S-K. Clustering of load profiles of residential customers using extreme points and demographic characteristics. Electronics 2021;10(3):290. http://dx.doi.org/10.3390/electronics10030290.

[53] Okereke GE, Bali MC, Okwueze CN, Ukekwe EC, Echezona SC, Ugwu CI. K-means clustering of electricity consumers using time-domain features from smart meter data. J Electr Syst Inf Technol 10 2023;2. http://dx.doi.org/10.1186/s43067-023-00068-3.

[54] Sabri M, El Hassouni M. Photovoltaic power forecasting with a long short-term memory autoencoder networks. Soft Comput 2023;27:10533–53. http://dx.doi.org/10.1007/s00500-023-08497-y.

[55] Yang Wangwang, Shi Jing, Li Shujian, Song Zhaofang, Zhang Zitong, Chen Zexu. A combined deep learning load forecasting model of single household resident user considering multi-time scale electricity consumption behavior. Appl Energy 2022;307:118197. http://dx.doi.org/10.1016/j.apenergy.2021.118197.

[56] Htun HH, Biehl M, Petkov N. Survey of feature selection and extraction techniques for stock market prediction. Financ Innov 2023;9:26. http://dx.doi.org/10.1186/s40854-022-00441-7.

[57] Fan Dongyan, Sun Hai, Yao Jun, Zhang Kai, Yan Xia, Sun Zhixue. Well production forecasting based on ARIMA-LSTM model considering manual operations. Energy 2021;220:119708. http://dx.doi.org/10.1016/j.energy.2020.119708.

[58] Pan Yan, Jin Xiuling, Li Yuliang, Chen Dalian, Zhou Juan. A study on the prediction of book borrowing based on ARIMA-SVR model. Procedia Comput Sci 2021;188:93–102. http://dx.doi.org/10.1016/j.procs.2021.05.057.

[59] Zhang J, Zhao J, Cheng G, Rouhani A, Chen X. Explainable multi-fidelity Bayesian neural network for distribution system state estimation. Appl Energy 2025;392:125972. http://dx.doi.org/10.1016/j.apenergy.2025.125972.

[60] Machlev R, Heistrene L, Perl M, Levy KY, Belikov J, Mannor S, Levron Y. Explainable artificial intelligence (XAI) techniques for energy and power systems: Review, challenges and opportunities. Energy AI 2022;9:100169. http://dx.doi.org/10.1016/j.egyai.2022.100169.

[61] Neubauer Alexander, Brandt Stefan, Kriegel Martin. Explainable multi-step heating load forecasting: using SHAP values and temporal attention mechanisms for enhanced interpretability. Energy AI 2025;20:100480. http://dx.doi.org/10.1016/j.egyai.2025.100480.

[62] Xu W, Liu Y, Fan X, Shen Z, Wu Q. Short-term wind power forecasting based on dual attention mechanism and gated recurrent unit neural network. Front Energy Res 2024;12:1346000. http://dx.doi.org/10.3389/fenrg.2024.1346000.

[63] Han Mingdong, Fan Lingyan. A short-term energy consumption forecasting method for attention mechanisms based on spatio-temporal deep learning. Comput Electr Eng 2024;114:109063. http://dx.doi.org/10.1016/j.compeleceng.2023.109063.

[64] Teleron Jerry I, Gonzales Shem L, Fajardo Arnel C. A modified energy demand forecasting model using hybrid CNN-LSTM with transformer for univariate time series. IJCESEN 2025;11(3). http://dx.doi.org/10.22399/ijcesen.2293.

[65] Su L, Zuo X, Li R, Wang X, Zhao H, Huang B. A systematic review for transformer-based long-term series forecasting. Artif Intell Rev 2025;58:80. http://dx.doi.org/10.1007/s10462-024-11044-2.

[66] Gao Jiaxin, Chen Yuntian, Hu Wenbo, Zhang Dongxiao. An adaptive deep-learning load forecasting framework by integrating transformer and domain knowledge. In: Advances in applied energy, Vol. 10, 2023, p. 100142. http://dx.doi.org/10.1016/j.adapen.2023.100142.

[67] Moosbrugger L, Seiler V, Wohlgenannt P, Hegenbart S, Ristov S, Eder E, Kepplinger P. Load forecasting for households and energy communities: are deep learning models worth the effort?. 2025, arXiv preprint arXiv:2501.05000, Available at: https://doi.org/10.48550/arXiv.2501.05000.

[68] Batra N, Gulati M, Singh A, Srivastava MB. It's different: Insights into home energy consumption in India. In: Proceedings of the 5th ACM workshop on embedded systems for energy-efficient buildings, ser. BuildSys'13. New York, NY, USA: Association for Computing Machinery; 2013, p. 1–8, [Online]. Available https://doi.org/10.1145/2528282.2528293.

[69] Lian Huihui, Ji Ying, Niu Menghan, Gu Jiefan, Xie Jingchao, Liu Jiaping. A hybrid load prediction method of office buildings based on physical simulation database and LightGBM algorithm. Appl Energy 2025;377(Part C):124620. http://dx.doi.org/10.1016/j.apenergy.2024.124620.

[70] Hasan Mahmudul, Mifta Zannatul, Papiya Sumaiya Janefar, Roy Paromita, Dey Pronay, Salsabil Nafisa Atia, Chowdhury Nahid-Ur-Rahman, Farrok Omar. A state-of-the-art comparative review of load forecasting methods: characteristics, perspectives, and applications. Energy Convers Manag: X 2025;26:100922. http://dx.doi.org/10.1016/j.ecmx.2025.100922.

[71] Satpathy Priya Ranjan, Ramachandaramurthy Vigna Kumaran. Artificial intelligence and machine learning for distributed energy resource management systems: applications, frameworks, and future directions. Appl Energy 2026;403(Part B):127109. http://dx.doi.org/10.1016/j.apenergy.2025.127109.

[72] Quan Steven Jige, Xue Yang, Li Chaosu. Nonlinearity in the relationships between urban form and residential energy use intensity. Appl Energy 2025;383:125344. http://dx.doi.org/10.1016/j.apenergy.2025.125344.

[73] Murray D, Stankovic L, Stankovic V. An electrical load measurements dataset of United Kingdom households from a two-year longitudinal study. Sci Data 2017;4:160122.

[74] Makonin S, Ellert B, Bajić IV, Popowich F. Electricity, water, and natural gas consumption of a residential house in Canada from 2012 to 2014. Sci Data 2016;3:160037.

[75] Shin S, Choi S, Kim J, Lee W, Beka A, Boardman K, et al. ENERTALK: A national-scale dataset of electricity consumption from Korean households. Sci Data 2019;6:190011.